THE COLD START PROBLEM

Other people are valuable

USER COUNT LEADERBOARD

Week 3. Update your numbers.

Sift	15
Vouch	44
QuestCity	-
Rekindled	-
Roam	-
Circles	-
FreeWeight	-
Recruitr	-
Misfit AI	8
Evidence of Life	13
Tripable	6
Std. Deviants	10

TEAM PRESENTATIONS

~10 min each (5 min present, 5 min feedback).

DATE	CLASS	TEAM 1	TEAM 2	TEAM 3
Thu, Apr 9 (today)	c21	Circles	Evidence of Life	QuestCity
Tue, Apr 14	c22	FreeWeight	Vouch	Rekindled
Thu, Apr 16	c23	Std Deviants	Misfit AI	Roam
Tue, Apr 21	c24	NoticeMe		
Thu, Apr 23	c25	Millionaires	Recruitr	Sift

Format

Show the product live. Walk through your data. Where are you stuck?

You Choose the Focus

Practice presenting a demo. Get feedback from peers on your product. You choose how to structure your 5 minutes based on these goals.

CIRCLES

EVIDENCE OF LIFE

QUESTCITY

NETWORK EFFECTS

WHAT IS A NETWORK EFFECT?

Every new user makes the product more valuable for every existing user.

NETWORK EFFECT

- More users = more value for *everyone*
- Creates **defensibility**
- WhatsApp, Airbnb, LinkedIn

NOT A NETWORK EFFECT

- **Virality** = speed of user acquisition
- **Scale** = cost advantages from volume
- JibJab went mega-viral. Zero network effects.

💡 NFX estimates network effects account for ~70% of value created by tech companies since 1994. They are the strongest form of competitive defensibility.

THREE LAWS OF NETWORK VALUE

Each law describes a different type of network. The value growth accelerates as you move from broadcast to group-forming.

BROADCAST

$$V = N$$

Sarnoff's Law

One central node reaches many. Value grows linearly.

Radio, TV, newsletters

→

COMMUNICATION

$$V = N^2$$

Metcalfe's Law

Every user can connect with every other. Value grows with the square.

Phone networks, email, social media

→

GROUP-FORMING

$$V = 2^N$$

Reed's Law

Groups can form freely. Value grows exponentially.

WhatsApp groups, Slack, Facebook groups

💡 **The implication:** If your product enables group formation (not just 1-to-1 connections), the potential value growth is exponential, not quadratic. But all three laws overestimate in practice.

WHY ALL THREE LAWS ARE WRONG

Useful as mental models, misleading as math. The actual relationship is closer to $N \cdot log(N)$.

WHY THEY OVERESTIMATE

- **Not all connections are equal.** Your 1,000th Facebook friend adds almost no value.
- **Cognitive limits.** Humans can maintain ~150 meaningful relationships (Dunbar's number).
- **Reed's Law at scale is absurd.** A network of 50,010 people cannot be 1,000x more valuable than 50,000.

WHAT THEY MISS ENTIRELY

- **Phase 0.** All three laws start at N > 0. They say nothing about the most dangerous phase: getting from zero to critical mass.
- **Engagement quality.** 1M dormant users have less value than 100K active ones.
- **User heterogeneity.** Real networks have buyers and sellers, creators and consumers.

💡 **The Allee Effect** (ecology) is a better model: below a threshold, networks collapse. Above it, they self-sustain. Beyond carrying capacity, growth *decreases* value.

EFLACTEM'S LAW

Metcalfe's Law in reverse. When users leave, value destruction is *disproportionate*.

Metcalfe says value = N^2

200 users → **100 users**

Value: 40,000 | Value: 10,000

50% user loss = 75% value loss

WHY THIS MATTERS

- Death spirals feel sudden but are mathematically predictable
- Network unraveling starts from the **most connected nodes**, not the periphery
- When "stars" leave, they take disproportionate value and trigger cascading departures
- Research on Friendster's collapse confirmed this pattern

NEGATIVE NETWORK EFFECTS

Networks don't automatically improve with scale. Past a certain point, growth actively destroys value.

CONGESTION

The network becomes overloaded.

- Highways at rush hour: more cars = slower for everyone
- Streaming during major events: servers overwhelmed
- Dating apps: most attractive users get flooded with messages and disengage

POLLUTION

Low-quality participants degrade the experience.

- Social feeds flooded with spam and bots
- Marketplaces with too many low-quality listings
- Group chats past ~20 people: conversation quality collapses

CARRYING CAPACITY IS REAL

- Food delivery: conversion plateaus at ~45 restaurants in a 7-min radius
- Uber: once wait times hit 4-5 min, more drivers add negligible value

HOW TO MANAGE IT

- Algorithmic filtering (News Feed over raw timeline)
- Trust systems (Airbnb reviews, eBay ratings)
- Access controls (privacy settings, workspace gating)

16 TYPES OF NETWORK EFFECTS

NFX's taxonomy, ranked by strength. Knowing your type tells you how durable your moat is.

CATEGORY	TYPES	EXAMPLES	STRENGTH
Direct	Physical, Protocol, Personal Utility, Personal, Market Networks	Telecom, Bitcoin, WhatsApp, Facebook, AngelList	**Strongest**
Two-Sided	Marketplace, Platform, Asymptotic Marketplace, Expertise	eBay, iOS, Uber, Salesforce	**Strong**
Data	Data	Waze, Yelp	**Moderate**
Tech Performance	Tech Performance	BitTorrent, Skype	**Moderate**
Social	Language, Belief, Bandwagon, Tribal	"Googling," Bitcoin, Slack hype, YC alumni	**Weakest**

💡 **Physical effects** (telecom infrastructure) are nearly impossible to displace. **Bandwagon effects** can evaporate overnight.

✅ **Most products layer multiple types.** LinkedIn combines personal, marketplace, expertise, and language effects. Each layer compounds defensibility.

TWO TYPES THAT MATTER FOR YOU

DIRECT NETWORK EFFECTS

More users of the **same type** = more value.

- Every phone user benefits from more phone users
- WhatsApp, iMessage, LinkedIn connections

INDIRECT NETWORK EFFECTS

More users of **one type** attracts a **different type**.

- More iOS users attracts more developers, which attracts more users
- More Uber drivers = better for riders (and vice versa)

⚠️ **Same-side negative effects are real.** More Uber drivers = less income per driver. More eBay sellers = more competition per seller. Growing one side can hurt that side.

LOCAL VS. GLOBAL NETWORK EFFECTS

This distinction matters more than people think.

LOCAL (UBER)

- A driver in NYC only helps riders in NYC
- Every new city starts from zero
- Uber's massive US base gave them zero advantage against Didi in China

GLOBAL (AIRBNB)

- A new host in Barcelona benefits travelers everywhere
- Competitors need global supply AND global demand simultaneously
- Result: Airbnb has few meaningful international competitors

💡 **For your teams:** If your network effects are local, your moat is local. Being #1 on one campus doesn't protect you on another. But if supply in one place benefits demand everywhere, scale compounds fast.

NETWORK EFFECTS AS A MOAT

NFX ranks the four ways a company becomes hard to kill. Network effects are the strongest.

1

Network Effects

More users = more value. Self-reinforcing.

~70% of tech value since 1994

2

Embedding

Switching costs. Integrations. Data lock-in.

Oracle, Salesforce, Workday

3

Scale

Cost advantages from volume. Economies of scale.

Amazon, Walmart, AWS

4

Brand

Psychological switching costs. Identity. Trust.

Apple, Nike, Patagonia

Most winning companies layer multiple defensibilities. But network effects are the one that compounds on its own.

THE COLD START PROBLEM

THE COLD START PROBLEM

Your product needs users to be valuable, but nobody wants to use a product with no users.

User arrives
Empty product

→

No value
User leaves

→

Still empty
Next user leaves too

→

Dead

- Before critical mass, the network **actively repels** users. This is the default state.
- You only get **one shot** at a first impression. Users who bounce don't come back to check later.
- The solution isn't "get more users." It's **get the right users, in the right density, in one place.**

FIVE TYPES OF COLD START

The type determines the strategy. Get the diagnosis right first.

🐔🥚

Network

Two groups need each other. Neither joins first.

Dating apps, ride-sharing

📪

Content / Supply

Platform works but has nothing on it.

Reddit, Yelp, YouTube

🏪

Marketplace

No buyers AND no sellers. Transactions are the value unit.

Airbnb, DoorDash, Etsy

👥

Group / Invite

Useless unless a whole group adopts together.

Slack, Zoom, multiplayer games

🤖

AI / Algorithm

Recommendations need data you don't have yet.

Spotify, TikTok, Netflix

💡 **Quick diagnostic:** Has two distinct user groups? Network/Marketplace. Empty without content? Content. Requires a whole group? Group/Invite. Needs user data for AI? AI/Algorithm.

THE ATOMIC NETWORK

The smallest group of users that can sustain itself without your constant intervention.

PRODUCT	ATOMIC NETWORK	SIZE
Zoom	2 people wanting to video call	**2**
Slack	1 company team	**~10**
Uber	~15 concurrent drivers near one transit hub	**~15**
Facebook	Harvard's undergraduate student body	**~6,700**
Airbnb	~300 listings + 100 reviews in one city	**~400**

⚠️ **The test:** Can you name specific people in your atomic network? If your answer is "anyone interested in X," you don't have one yet. Be concrete: "the 15 people in my dorm who use group chats to plan dinner."

EVERY NETWORK HAS A HARD SIDE

The user group hardest to recruit, whose presence makes the product valuable for everyone else.

PRODUCT	HARD SIDE	EASY SIDE
Uber	**Drivers**	Riders
YouTube	**Creators**	Viewers
Tinder	**Women**	Men
Wikipedia	**Editors (0.02%)**	Readers
Slack	**Workspace admins**	Team members

The 1/10/100 Rule: For every 100 users, 10 engage passively, 1 creates content. The 1% who create are the hard side. **All early resources go here.**

DENSITY BEATS GENERIC SCALE

50 users who all know each other > 500 users who don't.
A product that works perfectly for 50 people will spread.
A product that barely works for 5,000 will die.

SEVEN STRATEGIES THAT WORK

PICK TWO, NOT SEVEN

Successful companies used a median of 2 tactics to break cold start. Focus beats diversification.

1. Seed the Supply

Populate content yourself so it's not empty on arrival. You are the first 20 posts, listings, reviews.

Reddit, Yelp, GrubHub

2. Constrained Launch

Go small to feel big. 100% of a small network beats 1% of a large one.

Facebook, Tinder, Nextdoor

3. Single-Player Mode

Useful alone first. Come for the tool, stay for the network.

Instagram, OpenTable, Eventbrite

4. Invite Mechanics

The product requires or rewards inviting others. Exclusivity creates demand.

Dropbox, Gmail, PayPal

5. Wizard of Oz

Be the other side of the market yourself. Users see a working product; behind the scenes, it's you.

DoorDash, Zappos, Airbnb

6. Piggyback

Attach to a platform where your users already are. Use their distribution.

Airbnb+Craigslist, YouTube+MySpace

7. Event-Driven Launch

Use a time-bounded event to concentrate density in one moment.

Airbnb+DNC, Tinder+frat parties

1. SEED THE SUPPLY

Pre-populate the platform so users arrive to something real, not an empty room.

- **Reddit:** Founders created hundreds of fake accounts and posted content themselves for months
- **Yelp:** Created "Elite Squad," recruiting prolific local reviewers with exclusive events
- **GrubHub:** Manually scanned restaurant delivery menus and posted them before restaurants joined

FOR YOUR TEAM

- Write the first 20 reviews yourself
- Curate the first listings manually
- Scrape and import relevant content
- Don't launch social features until a new user sees immediate value

💡 If a user opens your product and it feels empty, you haven't seeded enough.

2. CONSTRAINED LAUNCH

Launch to the smallest self-sustaining network. 100% of a small network beats 1% of a large one.

- **Facebook:** Harvard only. 1,200 signups in 24 hours. Half of Harvard within a month. Then college by college.
- **Tinder:** USC sorority chapters first. Had women install on the spot, then walked to the frat house.
- **Slack:** One company team at a time. Teams that sent 2,000 messages had 93% retention.

FOR YOUR TEAM

- Pick one dorm, one club, one class, one company
- Get to density there before expanding
- Don't invite outside the group until it's clearly working inside it

⚠️ Columbia gives you natural atomic networks: a cohort, a floor, a club. Use them.

3. SINGLE-PLAYER MODE

Build value for user #1 with zero other users. Add the network layer later.

- **Instagram:** Photo filters were useful alone. 1M users in 3 months before network effects kicked in.
- **OpenTable:** Sold restaurant management software, not the dining network. Booking layer came later.
- **Eventbrite:** Free self-service event pages. 17% converted to paid ticketing within 12 months.

FOR YOUR TEAM

Ask yourself:

- "What is my product worth to user #1, who knows no one else on it?"
- If the answer is "nothing," add standalone utility or change strategies

Come for the tool.
Stay for the network.

4. INVITE MECHANICS

The product requires or rewards inviting others. Exclusivity creates demand.

- **Dropbox:** Both parties get 500MB free storage. 100K to 4M users in 15 months (3,900% growth).
- **Gmail:** Invite-only launch. Invites sold on eBay for up to $150. ~1M users through invite chains alone, zero paid acquisition.
- **PayPal:** $10 referral program + embeddable "Pay with PayPal" buttons on eBay listings. Every listing became a viral touchpoint.

FOR YOUR TEAM

- You can't afford cash incentives. Lean on **exclusivity**: closed beta, hard cap of 30 spots.
- Build **collaboration-required invites**: features that only work when you invite someone.

⚠️ Referral programs only work **after retention is proven.** Dropbox's product already worked. Fix retention before driving growth.

5. WIZARD OF OZ

Manually be the supply side. Users see a working product; behind the scenes, it's you.

- **DoorDash:** PDF menus + Google Voice number. Founders delivered the first 100 orders themselves.
- **Zappos:** Photographed shoes at retail stores, listed online, bought at retail when orders came in. Sold for $1.2B.
- **Airbnb:** Rented a $5K camera, photographed apartments door-to-door. NYC revenue doubled in one month.

FOR YOUR TEAM

- If you're a marketplace: **do the supply side yourself**
- If it's content curation: curate manually
- If it's a service: perform the service
- Track what breaks. The manual process reveals what to automate next.

💡 Your users don't care if the supply is automated or manual. They care if the product feels alive.

6. PIGGYBACK ON EXISTING NETWORKS

Attach to a platform where your users already are. Use their distribution.

- **Airbnb → Craigslist:** Reverse-engineered cross-posting. Listings looked more professional than native Craigslist and linked back to Airbnb.
- **YouTube → MySpace:** Built an embeddable video player for MySpace profiles. When MySpace tried to ban embeds, users revolted.
- **PayPal → eBay:** "Pay with PayPal" buttons on every listing. Every listing became a distribution channel.

FOR YOUR TEAM

Map your target users' existing platforms:

- Columbia Facebook groups
- Reddit r/Columbia
- Discord servers, WhatsApp groups
- LinkedIn

💡 Provide genuine value in those communities before pitching your product.

7. EVENT-DRIVEN LAUNCH

Use a time-bounded event to concentrate density in one moment.

- **Airbnb + DNC 2008:** 80,000 attendees, hotels booked solid. Targeted a moment of extreme housing demand.
- **Tinder + frat parties:** Download at the door = entry. Instant activated users with immediate social context.
- **Nextdoor + Halloween:** Launched 3 weeks before, when neighbors naturally interact and need to coordinate.

FOR YOUR TEAM

Pick a real event in the next 14 days:

- End-of-semester showcases
- Club events, hackathons
- Or create your own: a launch party with 30 people in a room

💡 30 people gathered for your launch > 30 cold users discovered over 3 weeks.

WHAT KILLS COLD START SOLUTIONS

BIG BANG LAUNCH

Releasing broadly before any dense community exists.

- **Google+** launched to hundreds of millions of potential users. 90% of sessions lasted under 5 seconds. Never built a single working atomic network.
- **Color** raised $41M and reached #2 in the App Store. First-time experience was a blank screen. No one nearby was using it, so the photo-sharing features had nothing to show.

WHY IT FAILS

- Broad launch = thin distribution everywhere, density nowhere
- Every user has a bad first experience (empty product)
- They leave and form a permanent negative opinion
- You burned your one shot at a first impression at scale

⚠️ The opposite of a big bang launch is a **constrained launch**. Go small. Go dense.

NOT FOCUSING LOCALES

Optimizing for total user count instead of connections between users.

- **Sidecar** launched in 20+ cities and achieved critical mass in none. They innovated ride-sharing before Uber and lost.
- **Uber** saturated one city at a time before moving to the next. Dominated every market they entered.

1,000,000 users with 2 connections each

<

100,000 users with 30 connections each

WHY IT FAILS

- Users in a sparse network don't encounter each other
- The product feels empty even with a large user count
- No individual market reaches the tipping point
- Resources spread thin across too many fronts

💡 Network value comes from **connections**, not users. Measure connections per user, not total users.

WRONG-SIDE SEEDING

Starting with demand before you have supply.

- **16 of 17** successful marketplaces in Lenny Rachitsky's research focused on **supply first**
- Demand that arrives to an empty marketplace leaves permanently and doesn't come back
- Supply often brings its own demand (creators share their pages, hosts list on multiple platforms)

WHY IT FAILS

- You drive traffic to your site. There's nothing there.
- Users bounce. They form a negative impression.
- You've now spent your acquisition budget on users who will never return
- The supply side hasn't seen demand, so they don't join either

✅ **The fix:** Build supply first. Always. Then let demand discover a product that already has something to offer.

DISCOUNTS TO SOLVE COLD START

Using price to attract users who wouldn't use the product at market rate.

- **Homejoy** offered $19 home cleanings (vs. $85 market rate)
- 75% of users were discount-seekers
- 25% month-1 retention
- Under 10% six-month retention
- Raised $40M. Dead.

WHY IT FAILS

- Discount users are not your real users
- They came for the deal, not the product
- When price normalizes, they leave
- You never learn if the product works because you're measuring price sensitivity, not product-market fit

⚠️ The users who respond to deep discounts are not the cohort that sustains a marketplace.

HARD SIDE NEGLECT

Failing to provide enough value to the users who make the product work for everyone else.

- **Clubhouse** exploded during COVID lockdowns. Oprah, Elon Musk, celebrity speakers.
- No creator monetization. No tools for speakers.
- The most valuable speakers left within months
- Daily active users fell 76% in 9 months

WHY IT FAILS

- The hard side creates the value everyone else consumes
- If they're not rewarded (money, tools, status), they leave
- When the hard side leaves, the easy side has no reason to stay
- Network effects work in reverse: losing stars triggers cascading departures

💡 **For your team:** Who is doing the work that makes your product valuable? What are you giving them in return?

REFERRALS BEFORE RETENTION

Trying to grow before proving the product retains anyone.

- **Dropbox's referral worked** because the product already had strong retention. The prompt triggered at 80% storage capacity (a moment of real pain).
- Users referred by other users retained better than users from paid channels
- The flywheel only spins when the product is already sticky

WHY IT FAILS

- Referrals bring users to a product that doesn't retain
- Those users churn fast and tell their friends it's not worth it
- You've now burned your referral channel AND created negative word-of-mouth
- Incentivized referrals to a broken product **accelerate** the death spiral

⚠️ **The order matters:** Retention first. Then referrals. Never the reverse.

MATCH YOUR STRATEGY TO YOUR PROBLEM

YOUR SITUATION	PRIMARY STRATEGY	SECONDARY STRATEGY
Empty platform, no content	**Content seeding**	Single-player mode
Marketplace, no supply	**Wizard of Oz**	Constrained launch
Network product, nobody to connect with	**Constrained launch**	Invite mechanics
Good product, no distribution	**Piggyback**	Event-driven launch
Need a whole group to adopt	**Invite mechanics**	Event-driven launch

💡 **Supply first, almost always.** 16 of 17 successful marketplaces in Lenny Rachitsky's research focused on supply before demand.

UPDATE YOUR GROWTH PLAN

UPDATE YOUR GROWTH STRATEGY DOC

Add a **Cold Start Strategy** section to your existing growth plan. Answer these four questions.

1

WHAT IS YOUR COLD START TYPE?

Network, Content/Supply, Marketplace, Group/Invite, AI/Algorithm, or none?

2

WHO IS YOUR HARD SIDE?

Name specific people. Not demographics. "The 12 people in my dorm who..." not "college students."

3

WHAT ARE YOUR TWO STRATEGIES?

Use the decision table. Pick a primary and a secondary.

4

WHAT CAN YOU DO THIS WEEK?

One concrete action for each strategy. Not "think about it." Do it.

DELIVERABLES TRACKER

Due today (Thu, Apr 9)

User testing round 2 (3-5 sessions, onboarding focus)
Growth strategy doc + 20 users
API metrics endpoint
Weekly update #1

New: Update your growth strategy doc with a Cold Start Strategy section

Cold start type, hard side, two strategies, this week's actions

Every Tuesday

Weekly update: what got done, what you're building next, and WHY you're building it

Due Thursday, April 16

Analytics Snapshot (screenshots + funnel + analysis)

THE 14-DAY PLAYBOOK

Feature freeze is April 23. Here's how to spend the time.

DAYS 1-3: PREPARE SUPPLY

- Define your atomic network (one specific group)
- Seed content or set up WoZ operations
- Identify your piggybacking target or event anchor

DAYS 4-7: FIRST 10 USERS

- Do things that don't scale: email personally, show up in person
- Get 10 active users in your atomic network
- Watch what they do. Fix what breaks.

DAYS 8-14: FIRST 50 USERS

- If Days 4-7 worked: replicate (one more group)
- If they didn't: diagnose why before growing
- Start invite mechanics only after retention is proven

⚠️ **The only test that matters:** Give your product to someone who doesn't know you and has no obligation to be nice. Watch what they do. Do they come back the next day?

FEATURE FREEZE: APRIL 23.

14 days. Everything after that is measurement and pitch prep.

THE COLD START PROBLEM

Density beats scale. Pick two strategies. Execute this week.